Binomial expansion

a) Binomial expansion

Example 3

Write down the 5th term, in ascending powers of x, in the expansion of $(1 + x)^7$.

5th term: $\binom{7}{4} 1^{7-4} x^4$ — 5th term is the $(r + 1)$th term so $r = 4$.

$= \frac{7!}{(7-4)!4!} x^4 = 35x^4$ — Use your calculator to get 35.

Example 4

Find the first three terms, in descending powers of x, in the expansion of $(4x - 3)^5$.

1st term $= 1(4x)^5 = 1024x^5$

2nd term $= \binom{5}{1}(4x)^4(-3)^1$ — The two indices add up to 5 for each term.

$= 5(256x^4)(-3) = -3840x^4$

3rd term $= \binom{5}{2}(4x)^3(-3)^2$ — The two indices add up to 5 for each term.

$= 10(64x^3)(9) = 5760x^3$

$(4x - 3)^5 = 1024x^5 - 3840x^4 + 5760x^3 + \ldots$

Example 5

Find the coefficient of x^2 in the expansion of $(2 + 3x)^7$.

3rd term $= \binom{7}{2}(2)^5(3x)^2$ — 1st term has no x, 2nd term has x, 3rd term has x^2.

$= 21(32)(9x^2) = 6048x^2$ — Remember to square 3 as well as x.

Thus the coefficient is 6048. — Do not put x^2 with the answer.

b) More complex expansion

Example 7

Find the first five terms, in descending powers of x, in the expansion of $(x-2)\left(x+\frac{3}{x}\right)^9$.

$$\left(x+\frac{3}{x}\right)^9 = (x)^9 + \binom{9}{1}(x)^8\left(\frac{3}{x}\right)^1 + \binom{9}{2}(x)^7\left(\frac{3}{x}\right)^2 + \ldots$$

First expand $\left(x+\frac{3}{x}\right)^9$.

$$= x^9 + 27x^7 + 324x^5 + \ldots$$

Note: We know that the highest power of x will be x^{10} from $x \times x^9$.
The final expansion will have x^{10} to x^6.
x^6 comes from $x \times x^5$.
Thus we only need to expand the 2nd bracket to the term in x^5.

$$\begin{aligned}(x-2)\left(x+\frac{3}{x}\right)^9 &= x(x^9 + 27x^7 + 324x^5 + \ldots) - 2(x^9 + 27x^7 + 324x^5 + \ldots)\\ &= x^{10} + 27x^8 + 324x^6 + \ldots - 2x^9 - 54x^7 - 648x^5 + \ldots\\ &= x^{10} - 2x^9 + 27x^8 - 54x^7 + 324x^6 + \ldots\end{aligned}$$

Example 8

Given that the coefficient of x^3 in the expansion of $(1 + ax + 2x^2)(2 - x)^7$ is 560, determine the value of a.

$$\begin{aligned}(2-x)^7 &= 2^7 + \binom{7}{1}(2)^6(-x) + \binom{7}{2}(2)^5(-x)^2 + \binom{7}{3}(2)^4(-x)^3 + \ldots\\ &= 128 - 448x + 672x^2 - 560x^3 + \ldots\end{aligned}$$

Terms in x^3 in the expansion of $(1 + ax + 2x^2)(2 - x)^7$

$$= 1(-560x^3) + ax(672x^2) + 2x^2(-448x)$$

Coefficient of $x^3 = -560 + 672a - 896 = 672a - 1456 = 560$

$$672a = 2016$$

$$a = 3$$

9. Find $(1 + x)^4 - (1 - x)^4$. Write your answer in ascending powers of x.

10. a) Find the first five terms in the expansion of $(2 + x)^6$ in ascending powers of x.
    b) Using your expansion, show that $(2.03)^6 = 69.9804$ correct to 4 decimal places.

11. Find the ratio of the coefficient of the x^3 term to the coefficient of the x^4 term in the expansion of $\left(2x + \frac{1}{2}\right)^6$.

12. Find the 5th term in the expansion of $\left(2 - \frac{1}{3x}\right)^{11}$ in ascending powers of x.

13. $(1 + kx)^8 = 1 + 12x + ax^2 + bx^3 + \dots$
    a) Find the value of k, the value of a and the value of b.
    b) Use your values of k, a and b to find the coefficient of the term in x^3 in the expansion of $(1 - x)(1 + kx)^8$.

14. a) Expand $(1 + 2x)^3$.
    b) Show that $(1 + 2x)^3 + (1 - 2x)^3 = 2(1 + 12x^2)$.
    c) Hence solve $(1 + 2x)^3 - (1 - 2x)^3 = 8$.

15. a) Write down the first three terms, in ascending powers of x, in the expansion of $(1 + ax)^n$ where $a \neq 0$ and $n > 2$.
    b) The coefficient of the x^2 term is half the coefficient of the term in x. Show that $n = \frac{1+a}{a}$.
    c) Find the coefficient of the x^2 term when $a = \frac{1}{5}$.

16. The first three terms, in ascending powers of x, in the expansion of $\left(1 + \frac{a}{x}\right)^n$ are $1 + \frac{24}{x} + \frac{252}{x^2}$.
    Find the value of a and the value of n.

17. a) Write down the binomial expansion of $(1 + x)^5$.
    b) Use your answer to **(a)** to express $(1 - \sqrt{2})^5$ in the form $p + q\sqrt{2}$ where p and q are integers.

18. Find the coefficient of x^2 in the expansion of $(2 - 3x + x^2)\left(1 - \frac{x}{2}\right)^{10}$.

19. $(2 - ax)^8 = 2^b - 5120x + cx^2 + \dots$
    Find the value of a, the value of b and the value of c.

20. $(1 + x)^4(1 + ax)^7 = 1 + 74x^2 + bx^2 + \dots$
    Find the value of a and the value of b.

Summary:

Binomial expansion

- The **binomial expansion** of $(1 + x)^n$ is $1 + nx + \frac{n(n+1)}{2!}x^2 + \frac{n(n+1)(n+2)}{3!}x^3 + \dots + x^n$ where n is a positive integer.
- The coefficient of the term in x^r = the $(r + 1)$th term $= \binom{n}{r} = \frac{n!}{(n-r)!r!} = {}^n\mathbf{C}_r$.
- $(x + y)^n = \binom{n}{0}x^n y^0 + \binom{n}{1}x^{n-1}y^1 + \binom{n}{2}x^{n-2}y^2 + \dots + \binom{n}{n-1}x^1 y^{n-1} + \binom{n}{n}x^0 y^n$

  This is known as the **binomial theorem** where n is a positive integer.